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"VTU Internship Report-2" ON "Data Science using Python"

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BACHELOR OF ENGINEERING IN COMPUTER SCIENCE AND ENGINEERING

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Department of Computer Science and Engineering
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CERTIFICATE

This is to certify that the Internship Report entitled "**Data Science using Python**" is a Bonafide work carried out by **AKASH S (1ME22CS006)** in partial fulfilment for the award of **Bachelor of Engineering** in Computer Science and Engineering of Visvesvaraya Technological University, Belagavi, during the year **2025–2026**. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report, deposited in the department library. This internship report has been approved as it satisfies the academic requirements in respect of internship work prescribed for the Bachelor of Engineering Degree.

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ABSTRACT

This report details a comprehensive 12-week project-based internship completed at KodNest (February 9, 2026 – May 9, 2026), focusing on 'Data Science Internship'. The objective of this programme was to bridge the academia-industry gap by transitioning engineering students from a theoretical mindset to a professional mentality. The technical phase involved creating high-performance Data science applications using Python, Pandas, NumPy, Scikit learn, and Matplotlib for data analysis, modeling, and visualization. A major highlight of the internship was integrating the Agentic AI for rapid prototyping and AI-grounded responses within the application. The program emphasized real-world Data Science workflows, including data collection, preprocessing, exploratory data analysis (EDA), and model building using Python libraries. It incorporated version control with Git and structured problem-solving approaches for data-driven decision-making. Beyond technical skills, the internship focused on the 'Campus to Corporate' (C2C) framework, developing professional communication, data storytelling, and analytical thinking through structured methods and real-world project execution. The internship effectively built the core competencies required for real-world software engineering and prepared the intern for a sustainable professional career.

Keywords: Data Science, Python, SQL, Pandas, NumPy, Scikit-learn, Data Analysis, Data Visualization, Exploratory Data Analysis (EDA), Git, API Integration, KodNest.

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CHAPTER 1 — INTRODUCTION

1.1 Background of the Project

The food service industry generates vast amounts of data through online platforms such as Zomato, where restaurants are continuously reviewed, rated, and categorized by users. However, this data is often unstructured and underutilized, making it difficult to extract meaningful insights about customer preferences, restaurant performance, and market trends.

This project — *Restaurant Data Analysis using the Zomato Dataset* — was developed to address this gap by transforming raw, messy data into structured and actionable information. Using Python and data analysis libraries such as Pandas, NumPy, and Matplotlib, the project focuses on cleaning, processing, and analyzing restaurant data to uncover patterns in ratings, cuisines, cost, and service features like online ordering and table booking.

The objective is to build a clear understanding of the restaurant ecosystem through data-driven insights, enabling better decision-making and highlighting trends within the food industry.

1.2 Problem Statement

The Zomato dataset contains a large volume of restaurant-related information, including ratings, cuisines, cost, location, and service features. However, this data presents several challenges:

- The dataset contains **missing, inconsistent, and noisy values**, making it unsuitable for direct analysis.
- Important attributes such as ratings and cost are often stored in **non-uniform formats**, requiring preprocessing.
- There is **no clear structure or organization** to easily extract meaningful insights.
- Relationships between variables such as **ratings, votes, cost, and services** are not immediately visible.
- Without proper analysis, it is difficult to identify **trends, popular cuisines, and customer preferences**.

1.3 Objectives

- To perform **data cleaning and preprocessing** on the Zomato dataset by handling missing values, duplicates, and inconsistencies.
- To standardize and transform raw data into a **structured format suitable for analysis**.
- To explore the dataset using **Exploratory Data Analysis (EDA)** techniques.
- To analyze key factors such as **ratings, votes, cost, location, and cuisines** to identify patterns and trends.
- To visualize insights using **Matplotlib and Seaborn** for better understanding of data relationships.
- To identify **popular cuisines, top-rated restaurants, and city-wise distribution**.
- To examine the impact of features like **online ordering and table booking** on restaurant ratings and popularity.
- To store the cleaned dataset in an **SQLite database** and perform query-based analysis (ELT process) and to generate **data-driven insights** that can help understand customer preferences and restaurant performance.

CHAPTER 2 — COMPANY OVERVIEW

2.1 About the Organization

KodNest Company is a specialized edtech and software ecosystem edtech company founded by **Mr. Akash Pandey**. The organization operates as a strategic bridge between academic learning environments and the professional software industry, focusing on outcome-driven skill development, structured problem-solving, and AI adaptation for engineering students.

2.2 Domain and Focus Area

Field	Details
Domain	Data Science using Python
Technology Focus	Python, Pandas, NumPy, Matplotlib, Seaborn, SQLite
Project Category	Data Analysis / Exploratory Data Analysis (EDA) / Data Cleaning & Visualization
Target Users	Data Analysts, Data Science Enthusiasts, Researchers, Business Analysts
Development Period	[1-04-2026] – [23-04-2026]

2.3 Tools and Technologies Used

Tool / Technology	Purpose
Python	Core programming language for data analysis
Pandas	Data cleaning, manipulation, and preprocessing
NumPy	Numerical operations and array handling
Matplotlib	Data visualization and plotting graphs
Seaborn	Advanced statistical data visualization
SQLite	Storing and querying cleaned data (ELT process)
Jupyter Notebook	Interactive environment for analysis and visualization
GitHub	Version control and project management

CHAPTER 3 — SYSTEM REQUIREMENTS

3.1 Hardware Requirements

Component	Specification
Processor	Intel Core i5 / AMD Ryzen 5 or higher
RAM	Minimum 8 GB (16 GB recommended for large datasets)
Storage	Minimum 5-10 GB free (for dataset, notebooks, and outputs)
System Type	64-bit operating system
Internet Connection	Required for downloading dataset (Kaggle) and libraries

3.2 Software Requirements

Software / Library	Version	Purpose
Python	3.9+	Core programming language for data analysis
Jupyter Notebook	Latest	Interactive environment for coding and visualization
Pandas	Latest	Data cleaning, manipulation, and preprocessing
NumPy	Latest	Numerical computations and array operations
Matplotlib	Latest	Data visualization and plotting
Seaborn	Latest	Statistical data visualization
SQLite3	Built-in	Database for storing and querying cleaned data
Kaggle	Latest	Source for Zomato dataset
GitHub	Latest	Version control and project hosting

CHAPTER 4 — METHODOLOGY AND ARCHITECTURE

4.1 Development Methodology — Agile Approach

The project was developed using a structured data analysis methodology, where raw data is progressively refined into meaningful insights through multiple stages. Each phase focuses on transforming unstructured data into a clean, analyzable format and extracting valuable patterns.

Phase	Focus	Deliverable
Phase 1	Data Collection	Imported Zomato dataset from Kaggle (CSV format)
Phase 2	Data Cleaning & Preprocessing	Handled missing values, removed duplicates, standardized columns, corrected data types
Phase 3	Exploratory Data Analysis (EDA)	Analyzed relationships between variables like ratings, cost, cuisines, and location
Phase 4	Data Visualization	Generated charts and plots (bar charts, histograms, heatmaps, scatter plots)
Phase 5	Data Storage & Querying (ELT)	Stored cleaned data in SQLite and performed SQL-based analysis

4.2 Application Architecture

The project follows a structured **data analysis pipeline architecture**, where each layer is responsible for transforming raw data into meaningful insights:

- **Data Source Layer** — The Zomato dataset (CSV) obtained from Kaggle serves as the raw input containing restaurant details such as ratings, cuisines, cost, and location.
- **Data Processing Layer** — Using Pandas and NumPy, the raw data is cleaned and pre-processed by handling missing values, removing duplicates, standardizing formats, and converting data types.
- **Analysis Layer** — Exploratory Data Analysis (EDA) is performed to identify patterns, relationships, and trends among variables like ratings, votes, cost, and services.
- **Visualization Layer** — Matplotlib and Seaborn are used to create visual representations such as bar charts, histograms, scatter plots, and heatmaps to better understand the data.
- **Data Storage Layer (ELT)** — The cleaned and transformed data is stored in a SQLite database, enabling efficient querying and further analysis using SQL.

4.3 Navigation Flow

The project follows a structured flow where data moves through different components, each responsible for a specific task in the analysis process.

Component	Purpose
Data Input	Load Zomato dataset (CSV) into the environment using Pandas
Data Cleaning	Handle missing values, remove duplicates, fix inconsistent formats
Data Transformation	Convert columns (ratings, cost, votes) into usable numerical formats
Exploratory Analysis	Analyze relationships between variables such as ratings, cuisines, cost, and location
Visualization	Generate charts (bar plots, histograms, scatter plots, heatmaps) to represent insights

CHAPTER 5 — CODE IMPLEMENTATION

5.1 Module Description

Module 1 — Data Collection

This module is responsible for acquiring the dataset. The Zomato restaurant dataset is imported from Kaggle in CSV format and loaded into the environment using Pandas for further processing.

Module 2 — Data Cleaning & Preprocessing

This module handles the refinement of raw data. It includes:

- Removing null values and duplicate records
- Standardizing inconsistent column formats
- Cleaning attributes like ratings, cost, and location
- Converting data types into usable numerical formats

The goal is to transform messy data into a clean and structured dataset ready for analysis.

Module 3 — Exploratory Data Analysis (EDA)

In this module, the cleaned data is explored to uncover patterns and relationships. Key analyses include:

- Distribution of restaurants across cities
- Popular cuisines and their frequency
- Relationship between ratings and votes (popularity)
- Impact of features like online ordering and table booking

Module 4 — Data Visualization

This module focuses on representing insights visually using Matplotlib and Seaborn. Various plots are generated, such as:

- Bar charts for category comparisons
- Histograms for distribution analysis
- Scatter plots for relationship exploration
- Heatmaps for correlation analysis

These visuals make complex data easier to interpret.

Module 5 — Data Storage & Querying (ELT)

The final module involves storing the cleaned dataset into a SQLite database. SQL queries are then used to:

- Identify top-rated restaurants
- Analyze cost trends across locations
- Find the most popular cuisines

This enables structured querying and efficient retrieval of insights.

5.2 Key Code Snippets

5.2.1 Data Cleaning and Preprocessing — Pandas

This snippet demonstrates how raw Zomato data is cleaned and prepared for analysis by handling missing values, duplicates, and inconsistent formats:

```
import pandas as pd

# Load dataset
df = pd.read_csv("zomato.csv")

# Remove duplicate records
df.drop_duplicates(inplace=True)

# Handle missing values
df.dropna(subset=['rate', 'approx_cost(for two people)'], inplace=True)

# Clean rating column (remove '/5' and convert to float)
df['rate'] = df['rate'].str.replace('/5', '')
df['rate'] = pd.to_numeric(df['rate'], errors='coerce')

# Clean cost column (remove commas and convert to int)
df['approx_cost(for two people)'] = df['approx_cost(for two people)'].str.replace(',', '')
df['approx_cost(for two people)'] = df['approx_cost(for two people)'].astype(float)

# Final cleaned dataset preview
print(df.head())
```

5.2.2 Data Visualization — Matplotlib & Seaborn

This snippet demonstrates how relationships between variables (such as ratings and votes) are visualized to uncover patterns and trends:

```
import matplotlib.pyplot as plt
import seaborn as sns

# Scatter plot: Ratings vs Votes (Popularity)
plt.figure(figsize=(8, 5))
sns.scatterplot(x='votes', y='rate', data=df)

plt.title("Ratings vs Votes (Popularity)")
plt.xlabel("Number of Votes")
plt.ylabel("Rating")
plt.show()
```

5.2.3 Correlation Heatmap — Feature Relationships

This snippet visualizes relationships between numerical features to understand how variables like ratings, votes, and cost influence each other:

```
import seaborn as sns
import matplotlib.pyplot as plt

# Select relevant numerical columns
corr = df[['rate', 'votes', 'approx_cost(for two people)']].corr()

# Plot heatmap
plt.figure(figsize=(6, 4))
sns.heatmap(corr, annot=True, cmap='coolwarm')

plt.title("Correlation Heatmap")
plt.show()
```

5.2.4 SQL Query — Top Rated Restaurants by City (ELT)

After loading cleaned data into SQLite, this query extracts top-performing restaurants based on ratings:

```
import sqlite3

conn = sqlite3.connect("zomato.db")

query = """
SELECT location, name, rate
FROM zomato
WHERE rate IS NOT NULL
ORDER BY rate DESC
LIMIT 10;
"""

top_restaurants = pd.read_sql(query, conn)
print(top_restaurants)
```

5.2.5 Data Transformation — Feature Engineering

This snippet shows how new insights are derived by transforming existing data:

```
# Create cost per person feature
df['cost_per_person'] = df['approx_cost(for two people)'] / 2

# Categorize restaurants based on rating
def rating_category(rating):
    if rating >= 4:
        return "Excellent"
    elif rating >= 3:
        return "Good"
    else:
        return "Average"

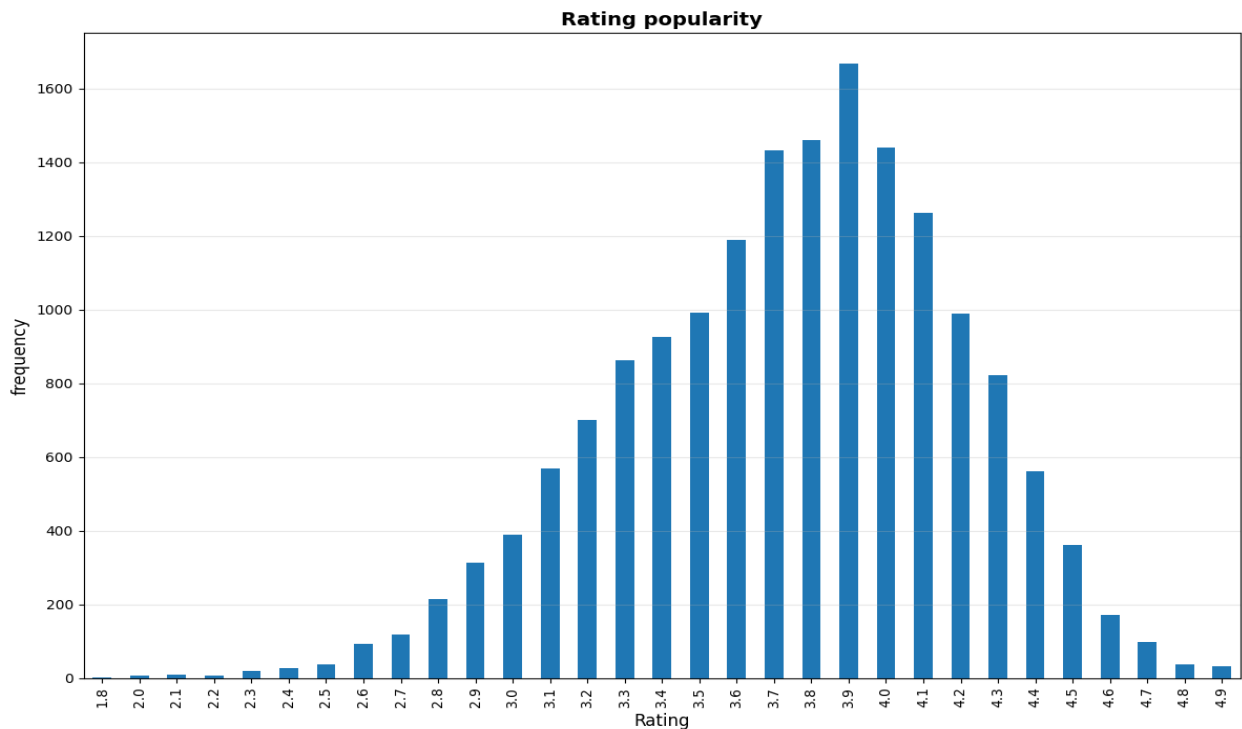
df['rating_category'] = df['rate'].apply(rating_category)

print(df[['rate', 'rating_category']].head())
```

CHAPTER 6 — EDA AND OUTPUT

6.1 Data Visualization:

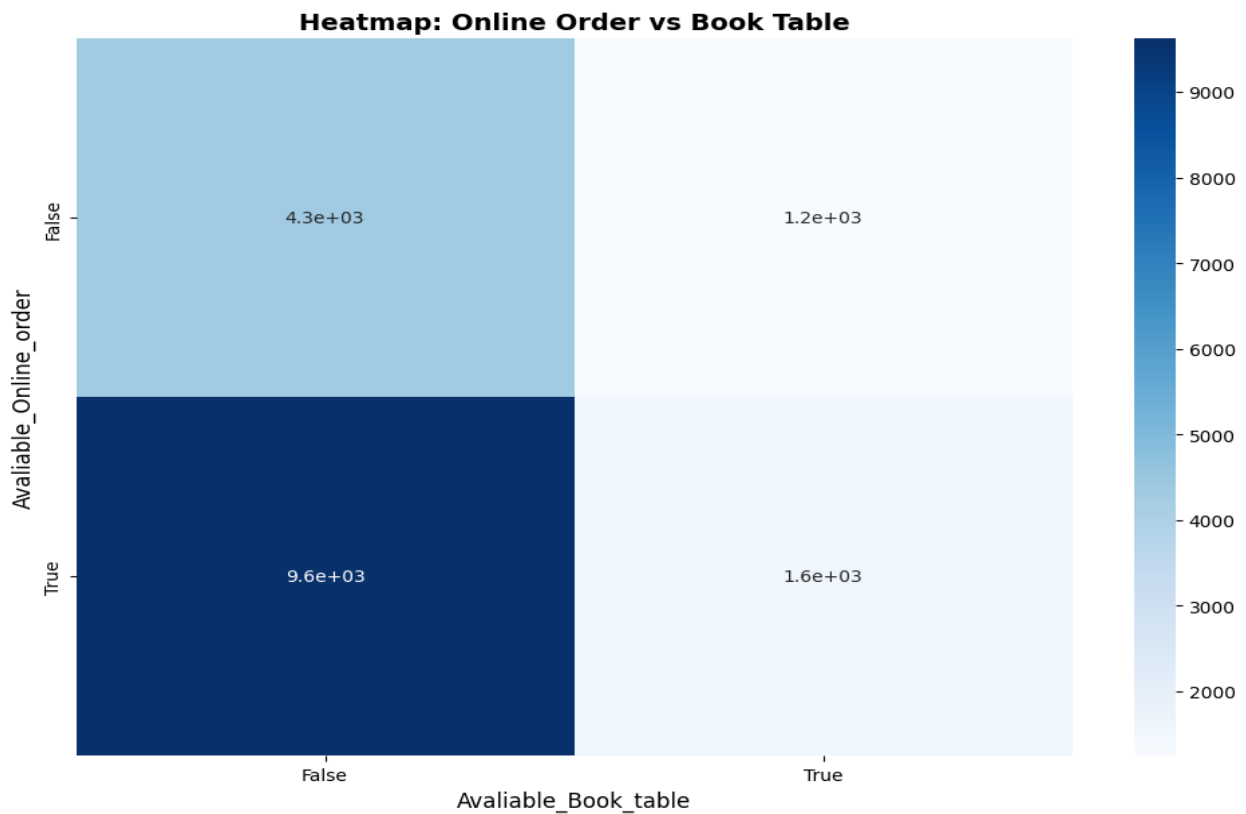
1. Rating popularity



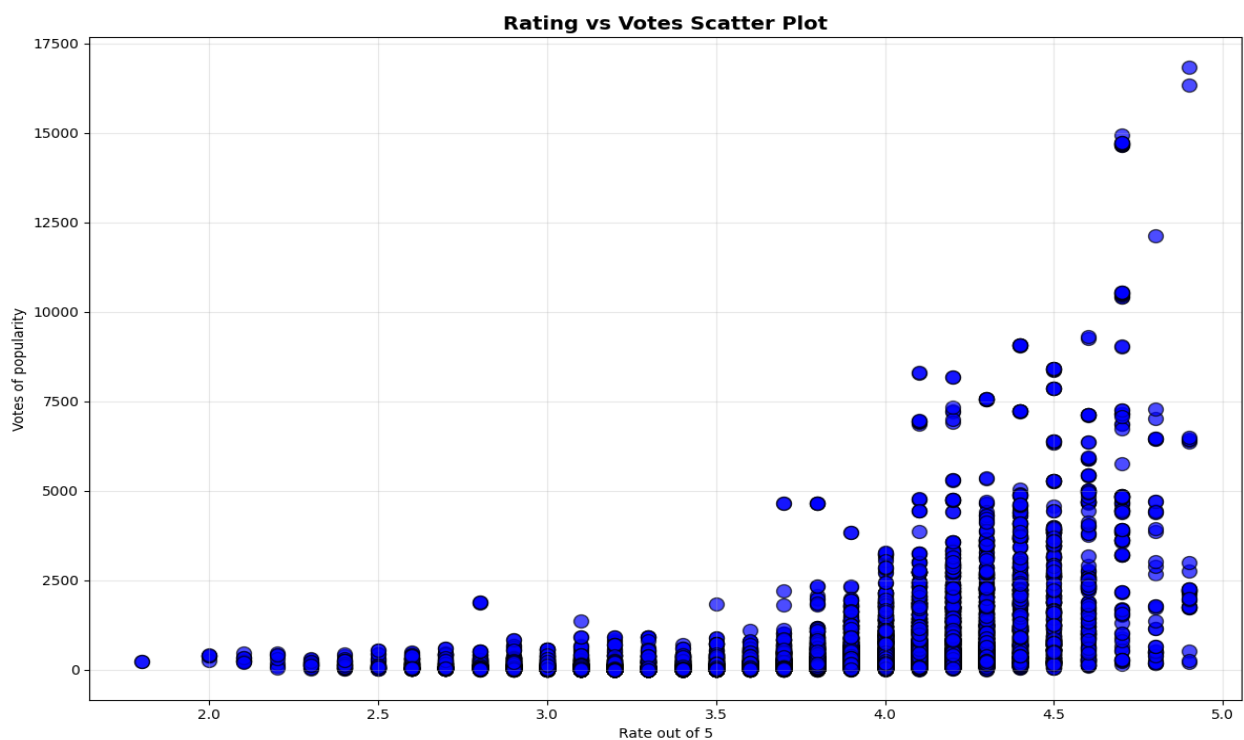
2. Online Order and Booking Table Comparison



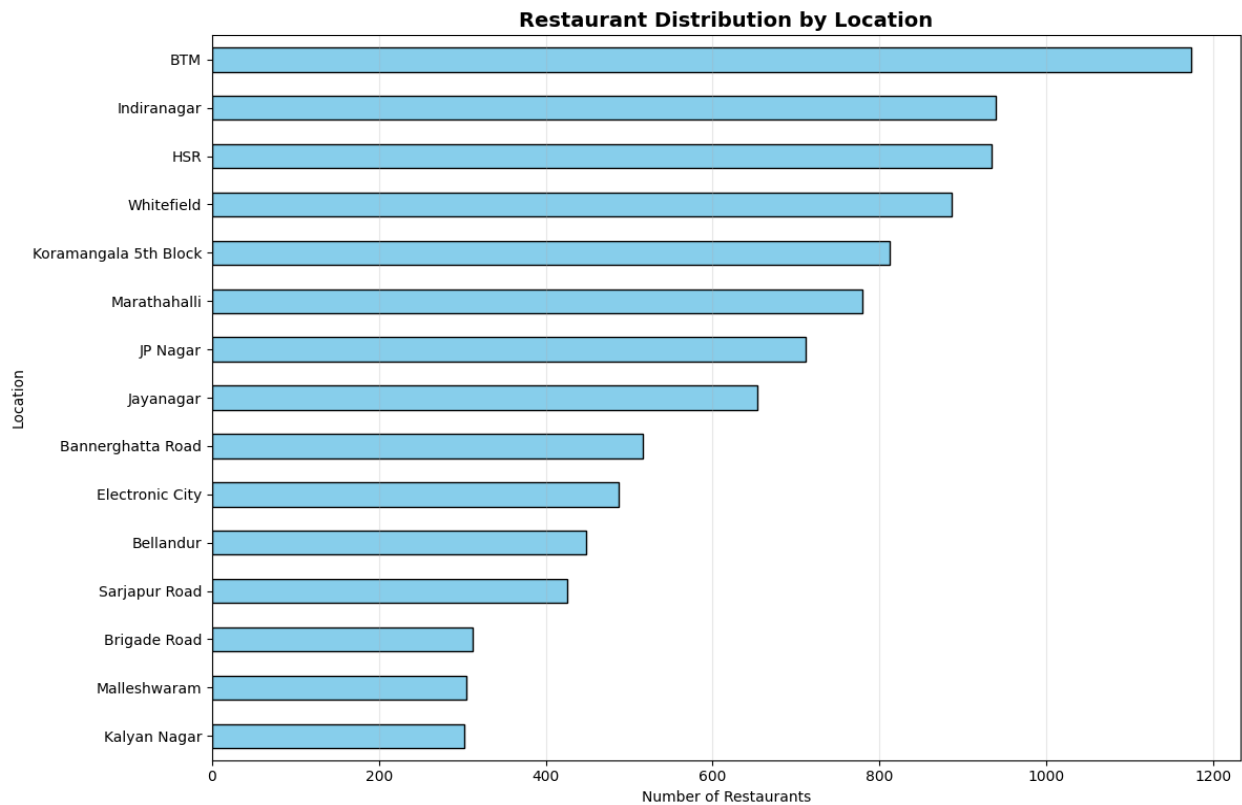
3. Online Order Vs Booking Table Using Heatmap



4. Rating vs Votes Scatter Plot



5. Restaurant Distribution by Location



6.2 Feature Verification

Feature	Status	Notes
Data Loading (CSV Import)	✔ Working	Zomato dataset successfully loaded using Pandas
Data Cleaning	✔ Working	Missing values, duplicates, and inconsistencies handled
Data Transformation	✔ Working	Ratings and cost columns converted to proper numerical format
Exploratory Data Analysis (EDA)	✔ Working	Patterns identified across ratings, votes, cuisines, and locations
Data Visualization	✔ Working	Charts generated using Matplotlib and Seaborn
Correlation Analysis	✔ Working	Relationships between variables identified using heatmaps
Feature Engineering	✔ Working	Derived columns like cost per person and rating categories
SQLite Integration (ELT)	✔ Working	Cleaned data stored and queried successfully
SQL Query Execution	✔ Working	Extracted insights like top-rated restaurants and popular cuisines
Insight Generation	✔ Working	Meaningful trends identified from dataset

CHAPTER 7 — CHALLENGES FACED

7.1 Technical Challenges

Challenge 1 — Handling Missing and Inconsistent Data

The dataset contained missing values, inconsistent formats (e.g., ratings like "4.2/5"), and irregular entries in multiple columns. This made direct analysis unreliable.

Resolution: Applied data cleaning techniques using Pandas such as removing null values, standardizing formats, and converting columns into proper numerical types.

Challenge 2 — Data Type Conversion Issues

Columns like ratings and cost were stored as strings with symbols ("/5", commas), preventing numerical analysis.

Resolution: Used string manipulation and type conversion methods (`str.replace`, `pd.to_numeric`) to transform them into usable numeric formats.

Challenge 3 — Large Dataset Handling

The dataset size and complexity made it difficult to process efficiently during analysis and visualization.

Resolution: Optimized operations using Pandas functions, filtered relevant columns, and avoided unnecessary computations to improve performance.

Challenge 4 — Visualizing Complex Relationships

Understanding relationships between multiple variables (ratings, votes, cost, cuisines) was not straightforward using raw data.

Resolution: Used Matplotlib and Seaborn to create scatter plots, bar charts, and heatmaps, making patterns and correlations easier to interpret.

Challenge 5 — Storing and Querying Cleaned Data

After cleaning and analysis, maintaining structured access to the processed data was challenging.

Resolution: Implemented an ELT process by storing the cleaned dataset in SQLite and executing SQL queries for efficient data retrieval and insight extraction.

7.2 Non-Technical Challenges

- Learning and applying data analysis concepts (**EDA, data cleaning, visualization**) while working on a real-world dataset.
- Understanding how to interpret data patterns and translate them into meaningful insights instead of just generating graphs.
- Selecting the right visualizations to represent different types of data effectively.
- Managing and organizing the workflow between **data cleaning, analysis, and visualization stages**.
- Ensuring the results are clear, concise, and understandable for non-technical audiences

CHAPTER 8 — CONCLUSION

The Restaurant Data Analysis using Zomato Dataset project was successfully completed, demonstrating a comprehensive workflow of data cleaning, exploratory data analysis (EDA), visualization, and data storage. The project showcases the practical use of Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn to transform raw, unstructured data into meaningful insights.

Through this project, the Zomato dataset was cleaned and standardized, enabling accurate analysis of key factors such as restaurant ratings, cost, cuisines, and location. Various visualizations and statistical techniques were used to uncover patterns, trends, and relationships within the data, providing a clear understanding of customer preferences and restaurant performance.

Additionally, the implementation of an ELT process using SQLite allowed efficient storage and querying of the cleaned dataset, further enhancing data accessibility and analysis capabilities.

This project highlights the importance of data preprocessing, visualization, and analytical thinking in solving real-world problems, and reflects skills that are essential for roles in data analysis and data engineering.

8.1 Learning Outcomes

- Gained hands-on experience in data cleaning and preprocessing using Pandas.
- Developed strong understanding of Exploratory Data Analysis (EDA) techniques.
- Learned to create effective data visualizations using Matplotlib and Seaborn.
- Understood how to handle real-world messy datasets and convert them into structured formats.
- Applied feature engineering to derive meaningful insights from existing data.
- Implemented an ELT pipeline using SQLite for data storage and querying.
- Improved ability to analyze patterns, trends, and relationships in data.
- Strengthened problem-solving skills in a data-driven environment.

CHAPTER 9 — FUTURE ENHANCEMENTS

The current version of the *Restaurant Data Analysis using Zomato Dataset* project establishes a strong foundation in data cleaning, analysis, and visualization. The following enhancements can further improve its capabilities:

9.1 Short-Term Enhancements (Version 2.0)

- Implement interactive visualizations using libraries like Plotly for better data exploration.
- Perform advanced statistical analysis to gain deeper insights into customer behavior.
- Add more feature engineering techniques to derive additional meaningful attributes.
- Improve data cleaning by handling outliers and extreme values more effectively.
- Automate the workflow using Python scripts or pipelines for repeatable analysis..

9.2 Long-Term Enhancements (Version 3.0)

- Build a dashboard (using Streamlit or Power BI) for real-time data visualization and interaction.
- Integrate machine learning models to predict restaurant ratings or customer preferences.
- Expand the dataset by combining multiple sources for more comprehensive analysis.
- Deploy the project as a web-based data application for wider accessibility.
- Implement a real-time data pipeline using tools like Apache Spark or Kafka.
- Add an AI-powered recommendation system for suggesting restaurants based on user preferences.

CHAPTER 10 — REFERENCES

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